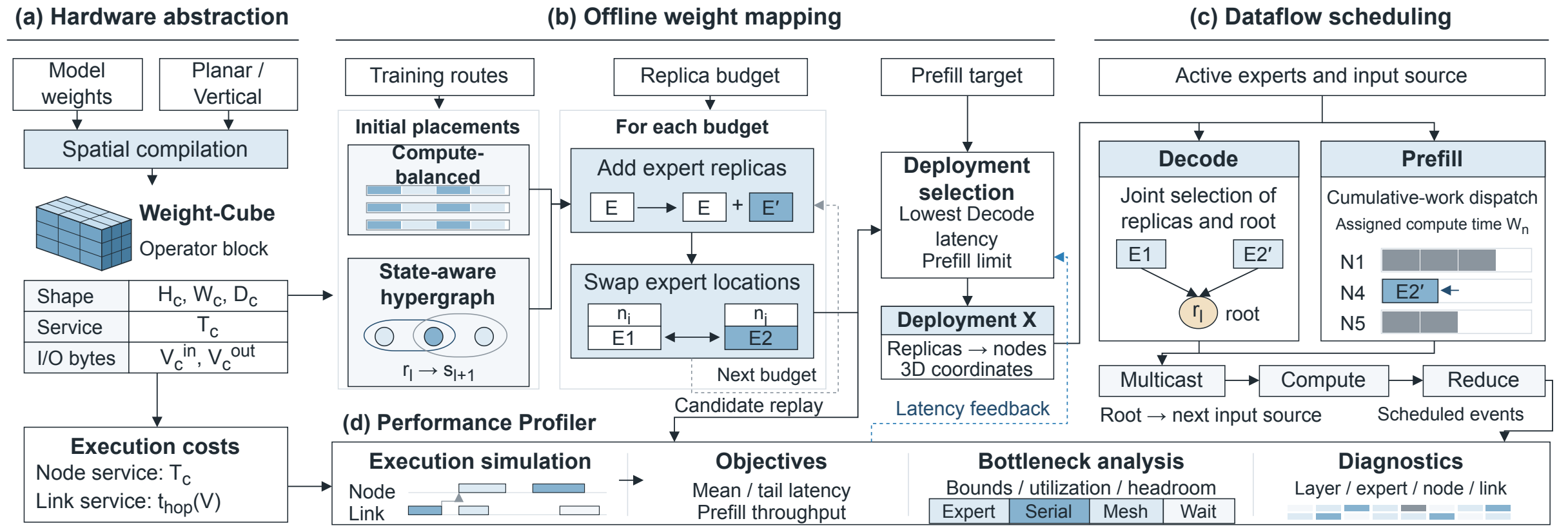

(a) Hardware abstraction
(b) Offline weight mapping
(c) Dataflow scheduling
Model weights
Planar / Vertical
Spatial compilation
Weight-Cube
Operator block
Shape
H_c, W_c, D_c
Service
T_c
I/O bytes
V_c^{in}, V_c^{out}
Execution costs
Node service: T_c
Link service: $t_{hop}(V)$
Training routes
Initial placements
Compute-balanced
State-aware hypergraph
$r_l \rightarrow s_{l+1}$
Replica budget
For each budget
Add expert replicas
E
E + E′
Swap expert locations
n_i
E1
E2
Next budget
Candidate replay
Prefill target
Deployment selection
Lowest Decode latency
Prefill limit
Deployment X
Replicas → nodes
3D coordinates
Latency feedback
Active experts and input source
Decode
Joint selection of replicas and root
E1
E2′
r_l root
Prefill
Cumulative-work dispatch
Assigned compute time W_n
N1
N4
E2′
N5
Multicast
Compute
Reduce
Root → next input source
Scheduled events
(d) Performance Profiler
Execution simulation
Node
Link
Objectives
Mean / tail latency
Prefill throughput
Bottleneck analysis
Bounds / utilization / headroom
Expert
Serial
Mesh
Wait
Diagnostics
Layer / expert / node / link